Questions an evaluator will ask, and where the system answers them

Each answer is short enough to say in thirty seconds, and each points to something that can be **shown** on the live system rather than asserted. Live: https://nyaysetu-project.vercel.app

A. The idea and its novelty

0. Show me a real case where this would have mattered. Pune, May 2024: in a fatal crash case, the minor driver's blood sample was allegedly swapped inside a hospital with his mother's blood. It was caught only because police had a tip-off, drew a second sample and compared DNA days later. With NyaySetu the seal serial is bound at collection; the substitute vial carries a different seal and is refused at laboratory intake the same day. *Show:* `/case-study` → Play; then "Try the swap yourself".

0a. A hash only works for files. Most evidence is physical. Correct, which is why physical exhibits are bound to the serial of their tamper-evident seal. The salted digest is anchored when sealed; each receiving party reads the serial; a different one is refused, anchored and alerted. *Show:* an exhibit page → "Tamper-evident seal"; laboratory intake asks for the serial.

1. Why blockchain? Why not a well-secured database with audit logs? Because the party that runs the database is the party whose handling of the evidence may be challenged. The defence cannot be asked to trust the prosecution's own log. A public ledger that no party in the case controls lets anyone check, without permission. Only 32-byte fingerprints go on chain; the readable record stays in PostgreSQL. *Show:* `/verify` in a private window, no account; the transaction link on Etherscan.

2. What exactly is new here? Hashing evidence is not new. Four things together: the fingerprint is computed on the officer's device before upload; verification needs no permission and no account; a failed check is written to the ledger permanently and blocks the hand-over; and evidence, summons and bail share one verifiable record. A file can even be verified without its reference. *Show:* capture screen showing the fingerprint before upload; `/verify` → "No reference? Use the file itself".

3. How is this different from e-Sakshya? e-Sakshya records search-and-seizure media for BNSS §105 and stores it with the government. NyaySetu provides independent, public verification and follows the item through malkhana, laboratory, prosecution and court, and covers summons and bail. It complements e-Sakshya: the fingerprint of each e-Sakshya recording can be anchored and re-checked at every later hand-over.

4. The Department of Justice already has "Nyaya Setu". That is a WhatsApp legal-information chatbot. This project is a record system for evidence, summons and bail; the purposes do not overlap.

B. Does it actually work?

5. Is it deployed, or is it a mock-up? Deployed and live, with real transactions on Ethereum Sepolia from three source-verified contracts. *Show:* the status line in the site header ("All services operational · Ethereum sepolia"); `/api/health`; the contracts on Etherscan; `/transparency`.

6. Show me tampering being caught. *Show:* `/transparency` → Tamper lab → "Use a sample" → "Change one bit": the fingerprint changes almost completely. Then `/verify/<reference>` with `docs/demo-kit/exhibit-original.png` (match) and `exhibit-altered.png`, which differs in one pixel (no match).

7. What happens when a tampered item reaches the laboratory? The laboratory re-hashes on arrival. A mismatch is anchored on chain, the custody transfer is refused, and alerts go to the prosecution and the court. It cannot be corrected quietly; a later genuine transfer is a separate record. *Show:* Alerts page; the item's state history showing "Integrity mismatch".

8. How do you know it works beyond a demo? 194 automated tests (53 contract, 132 API, 9 mobile), plus a browser test of all 44 pages for errors and accessibility. Measured: 100% of one-byte changes detected across 200 files; 100% of 258 unauthorised access attempts blocked; no items lost with 35% of chain writes failing. *Show:* `docs/metrics/METRICS.md`; `npm test` output.

C. Law and admissibility

8a. Your system generates a charge sheet. Is it filed when it is generated? No. A generated final report is a watermarked draft. A readiness engine checks it against the record and the live ledger; only a police officer can approve the latest version, and only when nothing blocks; approval freezes the bytes and their SHA-256. It becomes FILED only when the prosecutor or registry enters the court's receipt number. Certificates and seizure memos are always marked "UNSIGNED DRAFT" — the system never signs for a person. *Show:* case page → Final Report → Readiness report; Prepare version → Approve.

8b. Can I verify without trusting your server at all? Yes. On the passport page, "Check it without NyaySetu" makes your browser ask a public Ethereum node for the registration transaction, check the contract and event, and compare the committed digest — our server is not involved. If no public node answers, it says NOT VERIFIED.

9. Is a blockchain record admissible in court? The court relies on the BSA §63 certificate, which must state the hash value. NyaySetu generates that certificate from the record, with the digest anchored on chain and a QR code so the court can verify the paper itself. The ledger is supporting proof of integrity, not a replacement for the certificate. *Show:* Generate the §63 certificate PDF; scan its QR code with a phone.

10. Which provisions does it support? BNSS §105 (recorded search and seizure), BNSS §176(3) (forensic collection at the scene), BSA §63 (hash certificate), BNSS summons and bail provisions. *Show:* the "Why now" section on the home page.

11. Does a matching hash prove the evidence is genuine? No, and the system says so on its home page. It proves the file has not changed since it was anchored. How it was collected, and what it shows, remain matters for the court. GPS accuracy, device time and mocked-location flags help the court weigh it.

D. Privacy and security

11a. Under some State e-Sakshya rules the accused must pay to have a hash checked. Does that apply here? No. Verification is a public page and a public API: drop the file, get the ledger's answer, at no cost and without anyone's permission. Commentary on Madhya Pradesh's 2026 rules (LiveLaw, 11 Aug 2026) identifies exactly this cost as a fairness gap.

11b. Does the Supreme Court accept hash values? Yes. In *Pune Bar Association v. Union of India* (22 May 2026) the Court upheld BSA §63(4) and described a hash value as "synonymous with an electronic fingerprint". NyaySetu computes it at the scene and makes it checkable by all.

12. A public blockchain is permanent. What personal data is on it? None that identifies anyone. Case keys are salted hashes; no names, FIR numbers, addresses or file contents. Files are encrypted with AES-256-GCM and stored off chain. One exception, stated openly: bail check-in coordinates are on chain because the contract computes the geo-fence; moving that to a zero-knowledge proof is planned. *Show:* home page section "What goes on a public blockchain, and what never does"; a passport page showing no names or places.

13. Who can see what? Seven roles, each seeing only its part. Per-case assignment is enforced by the API and again by row-level security in the database. Defence access is requested and approved by the court. Every action and every refusal is logged. *Show:* Admin → Audit; Defence → Access requests.

14. What if someone steals an officer's account? Password plus an emailed one-time code; sessions are server-side and the registry can revoke them instantly; sensitive actions need two-person approval. Anything the attacker registers is attributed to that account on chain.

15. What if the officer's phone is compromised before hashing? Then the proof starts from the compromised file — no system can prove what happened before capture. Device time, GPS accuracy, clock skew and mocked location are recorded, and the anomaly screen flags implausible captures for review.

16. Is it compliant with the DPDP Act, 2023? Personal data stays in a database where it can be corrected or erased; the ledger holds only non-identifying fingerprints and keys, so erasure obligations do not collide with immutability.

E. Feasibility, cost and scale

17. What does it cost per record? Measured, not estimated: registering an item uses 3,60,093 gas; each custody transfer 1,56,407; scene to court about 8.3 lakh gas, independent of file size. On the public testnet this is free. On Ethereum mainnet it would be too expensive at state scale, so the recommendation for the

State is a permissioned EVM network (no gas fees) or a layer-2 network. *Show:* `/transparency` → "What a record costs"; `npm --prefix contracts run gas`.

18. Can it handle Maharashtra's volume? The API is stateless and scales horizontally; PostgreSQL handles lakhs of records a year comfortably. The ledger is the constraint, addressed by a permissioned network, by batching many fingerprints into one Merkle root, and by adding signers.

19. Police stations have poor connectivity. The field app queues captures on the phone with an idempotency key and syncs when signal returns, without duplicates. *Show:* aeroplane mode during capture, then reconnect.

20. Will officers actually use it? Capture is the same as taking a photo; the fingerprint is automatic. It removes work: the §63 certificate and custody record are generated instead of typed. The handbook explains every role's steps publicly.

21. How would it integrate with CCTNS, ICJS and e-Courts? Through its HTTP API. Any system can verify a file by posting only its fingerprint to the public lookup endpoint, and the case, summons and bail records can be exchanged with ICJS. The versioned API exists now at `/api/v1`, described in OpenAPI 3.1, with a sandbox client (`npm run sandbox:verify`). No government system is connected today; the exchange design, which needs each owning agency's approval, is in `docs/INTEGRATION.md`.

22. Who runs the ledger in production? Isn't a government-run chain just a database? Validator nodes held by institutions with different interests — police, judiciary, forensic directorate, a legal-services authority — so that no single party can rewrite the record. Fingerprints can additionally be checkpointed to a public network periodically.

F. Inclusion and impact

23. How does this help ordinary people? The accused sees obligations in Marathi, Hindi or English, has them read aloud, can give a reason for a missed check-in, and can verify evidence against them. Anyone can verify a file without an account. *Show:* Accused portal → "Listen"; switch language in the header.

24. Is it accessible? Large-text mode, light and dark themes, keyboard navigation, screen-reader labels; every page tested with axe-core. Installable on Android from the browser; works on low-end phones.

25. How will you measure impact? In a pilot: share of exhibits with a complete verified chain of custody; custody disputes in court; adjournments for service of summons; seizure-to-laboratory time; missed check-ins detected within a day; time to produce a §63 certificate.

G. Honesty and next steps

26. What does it not do yet? Citizen one-time codes use a sandbox provider (real Aadhaar needs a licensed agency); Hindi and Marathi text needs review by native speakers; the ledger is a testnet; device-signed captures are next.

27. What do you need to go further? A pilot partner — one police station, the regional forensic laboratory, one court and a legal-aid clinic — and guidance on integration with ICJS and on the State's preferred ledger infrastructure.